



Product, Experience, Data and Technical Architecture

A multi-tenant education operating system for institutions, academies, training providers and learning businesses

Prepared for Michael Ndhlovu
4 August 2026
Version 1.0

Document purpose and decision summary

This document defines the product model, institutional operating model, user journeys, permissions, information architecture, learning architecture, data architecture, integration strategy, technical stack, security posture, visual system and phased delivery plan for Veza Learning Cloud. It is intentionally designed as a licensable multi-tenant platform rather than a single-school implementation.

Decision	Selected direction	Why this is correct for Veza
Market identity	Veza Learning Cloud	“LMS” undersells the scope. Learning Cloud communicates a modular platform that includes administration, authoring, delivery, assessment, analytics and credentials. “Veza Studio” becomes the Canva-like authoring environment.
Commercial model	Multi-tenant SaaS with optional dedicated enterprise deployment	A shared control plane creates SaaS economics. A bridge model permits pooled tenants by default and siloed data or compute for regulated or high-value clients.
Product metaphor	Institution workspace + modular app suite	Each institution has a branded workspace, activated applications, shared data, unified search and consistent permissions. This is closer to Canva’s extensible workspace model than a conventional course list.
Primary architecture	Modular monolith first, event-driven boundaries, selective services	Education workflows are highly transactional and cross-domain. A modular monolith reduces distributed-system failure modes early while preserving bounded contexts that can be extracted when scale requires.
Cloud	AWS, South Africa region where service availability permits	Strong tenant-aware SaaS patterns, regional hosting, managed databases, object storage, media delivery, eventing and observability. Regional exceptions use documented data-residency controls.
Experience	Responsive web first, mobile learner app after core web maturity	Administration and content authoring require desktop density; learners require excellent mobile web immediately. Native/offline mobile follows once domain behaviour is stable.
Design direction	Professional bento minimalism	Dense institutional work is grouped into purposeful work surfaces and cards, without decorative card spam. Navigation is stable, contextual panels appear only where they support a task.

1. Product thesis

1.1 The problem Veza solves

Institutions do not merely publish courses. They recruit and enrol learners, define programmes and curricula, timetable cohorts, allocate staff, create learning material, deliver live and asynchronous teaching, assess performance, moderate results, manage interventions, communicate with guardians and employers, issue credentials, prove compliance and report outcomes. Most LMS products cover delivery but leave the institution to stitch together separate SIS, authoring, video, assessment, communication, document and analytics products.

Veza consolidates these workflows through a common institutional graph: people, organisations, academic structures, learning objects, schedules, evidence, outcomes and permissions. The product is not a monolithic screen. It is a coherent suite of applications sharing the same identity, data model, audit trail, design language and search layer.

1.2 Product pillars

Pillar	Promise	Architectural implication
Operate	Run the institution online	Organisation hierarchy, academic periods, admissions, enrolment, staff allocation, attendance, calendars, communication and records.
Create	Build rigorous learning experiences	Veza Studio, curriculum mapping, reusable blocks, templates, item banks, media, accessibility checks, versioning and approvals.
Teach	Deliver synchronous and asynchronous learning	Course rooms, live sessions, discussions, activities, assignments, adaptive release, offline-aware delivery and learner support.
Assess	Measure knowledge and performance credibly	Question banks, examinations, rubrics, practical evidence, moderation, accommodations, integrity controls and gradebook.
Understand	Turn activity into action	Progress, mastery, engagement, attendance, risk, cohort comparison, intervention workflows and institution-level BI.
Recognise	Issue portable evidence of achievement	Certificates, transcripts, Open Badges, comprehensive learner records, verification and expiry/renewal.
Extend	Connect the institution ecosystem	LTI, OneRoster, QTI, Common Cartridge, SSO, SCIM, APIs, webhooks, data exports and marketplace integrations.

1.3 Target institution segments

Segment	Distinct operating needs	Veza configuration
---------	--------------------------	--------------------

Schools and tutoring organisations	Terms, grades, subjects, classes, guardians, homework, attendance, report cycles	Guardian portal, class-based rostering, teacher planning, term reporting and age-sensitive consent.
Higher education and TVET	Faculties, departments, programmes, modules, semesters, credits, moderation, academic records	Programme/module hierarchy, credit rules, outcomes mapping, assessment boards and registrar workflows.
Corporate academies	Teams, job roles, mandatory training, recertification, manager visibility	Organisation-unit assignment, compliance paths, recurring certification and manager dashboards.
Professional training providers	Public catalogue, paid enrolment, cohorts, facilitators, certificates, CPD	Commerce, cohort operations, CPD points, verification and client-company reporting.
NGOs and public programmes	Low bandwidth, field cohorts, multiple languages, donor reporting	Offline-friendly delivery, lightweight media, facilitator mode, outcome reporting and regional hierarchy.
Content businesses	Reusable content, subscriptions, bundles, affiliates, branded academies	Multi-catalogue commerce, licensing rules, templates and storefronts.

2. Commercial and tenancy architecture

2.1 Tenant model

A tenant represents the contractual customer and top-level isolation boundary. A tenant can own one or more institutions, campuses, brands or academies. Users may belong to multiple tenants, but every authenticated request must carry an explicit tenant context. The tenant identifier is never accepted from an untrusted body field alone; it is derived from the authenticated session and authorised membership.

Layer	Entity	Purpose
Control plane	Veza operator organisation	Plans, entitlements, provisioning, billing, support, global configuration, feature flags, marketplace and operational analytics.
Tenant	Customer account	Contract, data residency, identity configuration, branding, domains, enabled modules, limits and retention policy.
Institution	Academic/legal unit	Institution profile, accreditation, calendars, campuses, departments and policies.
Organisation unit	Faculty, school, department, branch, company team	Delegated administration, reporting hierarchy, assignment and scoped permissions.
Learning space	Programme, course, module, subject, class, cohort	The operational context in which teaching and learning occurs.

2.2 Deployment tiers

Tier	Isolation	Commercial fit	Notes
Shared	Pooled application and database with tenant-keyed rows and database row-level controls	Small and mid-sized institutions	Lowest cost; strict tenant-aware access, quotas and noisy-neighbour controls.
Protected	Shared application, dedicated database/schema and optional dedicated media keys	Larger institutions and sensitive programmes	Stronger data isolation without duplicating the full platform.
Sovereign	Dedicated stack, database, storage, keys and network boundary	Government, defence, highly regulated enterprises	Higher cost; contract-specific release windows and infrastructure.

The product must preserve one operational model across tiers. Dedicated infrastructure must remain provisioned, upgraded and observed through the same control plane; otherwise the business drifts into bespoke managed hosting and loses SaaS leverage.

2.3 Packaging and licensing

Package dimension	Recommended mechanism
Base subscription	Institution fee including core administration, identity, calendar, course delivery, communication and reporting.
Active users	Monthly active learner bands rather than purchased seats for education providers with seasonal enrolment.
Modules	Studio Pro, Exams, Commerce, Advanced Analytics, Credentials, Guardian Portal, AI Assist, API/Integration Hub.
Consumption	Video processing/storage, live-session minutes, AI tokens, SMS/WhatsApp, proctoring and archival storage.
Enterprise options	Dedicated deployment, custom retention, premium support, sandbox tenant, migration services and certification.
Marketplace	Revenue share on paid templates, content packs, integrations and expert services.

3. Institutional domain model

3.1 Academic structure

The structure must avoid hard-coding a school-only or university-only vocabulary. Veza uses neutral canonical entities and lets each tenant relabel them. For example, “learning offering” can display as Course, Module, Subject, Unit or Programme according to the tenant taxonomy.

Canonical entity	Examples	Key rules
Academic period	Year, semester, trimester, term, intake	Nested periods; opening/closing, teaching, assessment and reporting windows are separately modelled.
Programme	BCom, Grade 10, Leadership Academy	Owns requirements, outcomes, credit rules, pathways and completion logic.
Learning offering	Module, subject, short course	Versioned curriculum definition independent from a delivery instance.
Course run	2027 Semester 1 Accounting 101	Concrete delivery instance with dates, staff, cohort, timetable and gradebook.
Class/section	Class A, evening group, Johannesburg cohort	Sub-group of a course run for teaching, scheduling or reporting.
Learning path	Onboarding path, qualification route	Ordered/conditional set of offerings, activities or credentials.
Competency/outcome	Learning outcome, unit standard, skill	Reusable framework nodes with alignment, evidence and mastery thresholds.

3.2 Course lifecycle

State	Owner	Entry criteria	Allowed actions	Exit control
Draft	Author/course designer	Course shell created	Build structure, map outcomes, add resources and assessments	Submit for review
Review	Reviewer/academic lead	Required metadata and checks pass	Comment, request changes, approve sections	Approve or return
Approved	Academic approver	Review resolved	Schedule publication, create course runs	Publish
Published	Course manager	Dates and learner access rules valid	Deliver, communicate, collect evidence	Close teaching
Marking	Assessors/moderators	Submission windows closed or rolling marking active	Mark, moderate, resolve exceptions	Ratify grades
Completed	Registrar/course manager	Completion rules evaluated	Issue results and credentials	Archive
Archived	Records manager	Retention package produced	Read, export, legal hold	Destroy only under policy

4. Users, identity and RBAC

4.1 User types

Persona	Primary jobs	Experience priority
Platform operator	Provision tenants, manage plans, global integrations, support and service health	Cross-tenant visibility must be exceptional, audited and time-bound. No routine access to learner content.
Tenant owner/executive	Set governance, approve configuration, view institution outcomes and risk	Executive overview, policy controls and delegated administration.
Registrar/operations administrator	Manage structures, periods, enrolment, records, timetables and progression	High-density operational tables, bulk actions, validation and exception queues.
Academic lead/curriculum manager	Design programmes, approve courses, map outcomes, assure quality	Curriculum graph, review workflows, coverage and quality checks.
Instructor/facilitator	Plan, teach, communicate, take attendance, assess and intervene	Today-focused workspace, class context, rapid actions and minimal administrative friction.
Assessor/moderator	Evaluate evidence and ensure fairness and consistency	Anonymous marking options, rubrics, sampling, variance and decision audit.
Learner	Understand what to do, learn, submit, track progress and obtain help	Mobile-first, low cognitive load, clear next action, resilient media and accessibility.
Guardian/sponsor/manager	Monitor authorised learner progress and respond to concerns	Consent-bound summaries and actionable alerts, not unrestricted course access.
External reviewer/accreditor	Inspect selected evidence, curriculum and reports	Time-limited, read-only evidence rooms with watermarking and complete audit.

4.2 Authorisation model: RBAC + scope + policy conditions

Pure role-based access control is insufficient because the same instructor may teach one course, moderate another and have no access to a third. Veza therefore uses a three-part decision: permission, resource scope and policy condition. Roles are bundles of permissions; assignments bind a role to a scope; policies constrain the action based on context.

Component	Example
Permission	assessment.grade, course.publish, learner.support.view, attendance.override
Scope	Tenant, institution, organisation unit, programme, course run, class, individual learner or owned resource
Condition	Only assigned assessor; only during marking window; only after conflict-of-interest declaration; only with step-up MFA;

	only fields allowed by guardian consent
Decision	Allow when the subject has the permission at a scope containing the resource and every applicable condition passes

4.3 Baseline role catalogue

Role	Representative permissions	Default scope
Tenant Owner	All tenant configuration, billing, policy and delegation; cannot silently impersonate users	Tenant
Institution Admin	Structures, users, enrolments, calendars, reports, integrations allowed by tenant policy	Institution
Registrar	Academic records, enrolment, progression, results release, transcript and retention	Institution/programme
Curriculum Manager	Programme design, outcomes, course templates, review and approval	Institution/department
Course Manager	Create runs, allocate staff, schedule, publish and close course	Course offering/run
Instructor	Deliver content, communicate, attendance, formative assessment, learner support	Assigned course run/class
Assessor	View allocated submissions and grade using authorised instruments	Assessment allocation
Moderator	Sample/review marks, return decisions and approve moderation	Assessment/course run
Learner	Access enrolled spaces, submit work, participate and view authorised records	Self + enrolments
Guardian/Sponsor	View consented learner summaries and communications	Specific learner relationship
Auditor	Read selected records and audit evidence	Explicit evidence room
Support Agent	Metadata and technical diagnostics; content access only through approved elevation	Tenant/support case

4.4 Sensitive access controls

- Impersonation is replaced by “assisted session”: explicit reason, ticket reference, user-visible banner, short expiry, prohibited actions, immutable audit and optional tenant approval.

- Grade changes after publication require a reason code, before/after values, actor, approver where policy requires, and learner notification rules.
- Health, disability, accommodation and disciplinary data use separate permissions and field-level encryption; instructors receive only the accommodation instructions necessary to deliver teaching.
- Bulk export is a privileged action with step-up authentication, purpose declaration, rate limits, watermarking and downloadable audit receipts.
- Role assignment cannot exceed the assigner's delegated authority; no administrator may grant a permission they do not possess at an equal or broader scope.

5. End-to-end user journeys

5.1 Institution onboarding

1. Sales creates prospect and selects deployment tier.
2. Authorised signatory accepts order, DPA and data-residency settings.
3. Control plane provisions tenant, default policies, encryption context, storage namespace and admin invitation.
4. Tenant owner configures branding, domain, terminology, academic calendar, authentication and required modules.
5. Setup wizard validates mandatory controls and creates a sandbox course.
6. Institution imports users/rosters or connects SSO/SCIM/OneRoster.
7. Readiness checklist runs: identity, email, privacy, grading, attendance, data retention, support contacts and integrations.
8. Tenant owner signs go-live approval; audit snapshot is retained.

5.2 Learner admission and enrolment

1. Applicant submits form or is imported from SIS/CRM.
2. Identity matching checks duplicates without exposing unrelated records.
3. Admissions workflow collects evidence, consent, prerequisites and payment where relevant.
4. Decision is approved, waitlisted or rejected with templated communication.
5. Accepted applicant activates identity or links institutional SSO.
6. Enrolment creates programme/course memberships, timetable, finance status, learner profile and required orientation path.
7. A rules engine detects timetable clashes, prerequisite failures, capacity limits and missing documents before confirmation.

5.3 Course creation and approval

1. Course designer selects an institution template or imports Common Cartridge.
2. Studio creates a versioned course blueprint with outcomes, modules, learning activities and assessments.
3. Coverage view identifies outcomes with insufficient learning or evidence.
4. Accessibility and quality checks identify heading, contrast, caption, alternative-text, broken-link, reading-load and assessment issues.
5. Reviewer comments on specific blocks; author resolves or records a justified exception.
6. Academic approver signs the immutable version.
7. Course manager creates a course run from the approved blueprint; local edits are overlays, preserving the source version and upgrade path.

5.4 Daily teaching

1. Instructor lands on Today: sessions, attendance, submissions, messages, learners requiring support and preparation tasks.
2. Opening a class switches the entire workspace into that class context.
3. Instructor launches lesson plan, live session, presentation, poll or activity from one teaching rail.
4. Attendance can be manual, code/QR, integration-assisted or evidence-based, but automated signals never silently become an official attendance record.
5. After class, the system prompts for notes, resources, attendance exceptions, follow-up activities and learner concerns.
6. Learner risk events enter an intervention queue with owner, due date, action and outcome.

5.5 Assignment and marking

1. Instructor creates task, rubric, outcomes, attempts, dates, late policy, originality policy and accommodations.
2. Learner sees requirements, estimated effort, rubric, submission formats and a pre-submit checklist.
3. Submission is virus-scanned, checksummed, timestamped and stored as an immutable attempt; subsequent attempts are separate versions.
4. Allocation supports named, anonymous, double or panel marking.
5. Marker grades against rubric, leaves anchored feedback and records confidence/flags.
6. Moderation samples by risk and policy; discrepancies trigger reconciliation.
7. Results release can be immediate, scheduled or gated by moderation; changes after release use the controlled correction flow.

5.6 Examination

1. Exam office selects secure assessment mode and candidate roster.
2. System validates identity method, accommodations, time zones, device requirements, question exposure rules and contingency plan.
3. Question assembly uses blueprint constraints, pools and randomisation while preserving equivalence.
4. Candidate performs system check and receives a practice environment.
5. During delivery, autosave, reconnect, incident capture and time adjustments are auditable.
6. Submission closes with a receipt. Incidents are reviewed separately from academic decisions.
7. Marking, moderation, ratification, release and appeals are distinct states with segregation of duties.

5.7 At-risk learner intervention

1. Analytics produces a signal with factors, confidence and freshness, not a hidden verdict.
2. Policy determines whether it is informational or creates a case.
3. Authorised staff review context and exclude known explanations.
4. Case owner contacts learner, records action plan and consent-sensitive notes.
5. Tasks and follow-up dates are assigned; unresolved cases escalate.
6. Outcome is recorded so the institution can evaluate intervention effectiveness and model bias.

6. Application suite and detailed capabilities

6.1 Home and command centre

Role-aware launchpad. It combines today's obligations, exceptions, institutional announcements, recent work and recommended next actions. Bento cards are prioritised by urgency and role, not used as decoration. Every card supports a direct task and can be dismissed, delegated or opened in context.

6.2 Institution Administration

Organisation hierarchy, campuses, academic calendars, terminology, policies, branding, domains, user lifecycle, cohorts, rooms, resources, custom fields, imports, retention and integration settings. Bulk operations always preview validation errors and produce reversible job logs where reversal is safe.

6.3 Admissions and Enrolment

Configurable applications, document collection, eligibility rules, review panels, offers, waitlists, payments, identity activation, programme registration, course selection and change requests. Workflow definitions are versioned so an applicant remains governed by the rules active when the application started.

6.4 Curriculum and Programme Designer

Visual curriculum graph linking programmes, offerings, outcomes, credits, prerequisites, co-requisites, assessment coverage and progression. Detects dead ends, circular prerequisites, overloaded periods and unmapped outcomes.

6.5 Veza Studio

Canva-like authoring workspace with a block canvas, outline, asset panel, properties panel, reusable components, institution templates, comments, version history, localisation, accessibility checks and approval. Supported blocks include text, callout, image, gallery, audio, video, embed, file, interactive, discussion, poll, quiz, assignment, live session, branching scenario, checklist and reflection.

6.6 Course Room

Learner-facing delivery space organised around Start/Continue, modules, calendar, discussions, people, grades and help. The default view emphasises the next meaningful action rather than a file repository.

6.7 Lesson Planning

Objectives, prior knowledge, teaching sequence, materials, differentiation, checks for understanding, homework and reflection. Plans can instantiate live teaching mode and later capture what actually occurred.

6.8 Calendar and Timetabling

Institution, programme, course, personal and external calendars. Supports recurring events, rooms, staff availability, capacity, time zones, clash detection, substitutions, assessment windows, office hours and iCalendar feeds. Timetable changes notify only affected audiences and preserve prior versions.

6.9 Live Learning

Provider abstraction for Zoom, Microsoft Teams, Google Meet, BigBlueButton or future providers. Veza owns scheduling, attendance policy, recording consent, metadata, links and post-session workflow; the provider handles conferencing. Recordings enter a processing pipeline for transcription, captions, chapters, access rules and retention.

6.10 Communication and Community

Announcements, direct messages, course discussions, channels, Q&A, group spaces, notifications and digests. Institutions control learner-to-learner messaging, guardian visibility, moderation, retention and quiet hours. Critical notices require acknowledgement tracking.

6.11 Attendance

Session roster, statuses, reasons, evidence, late/early thresholds, bulk capture, learner self-check-in, instructor verification, corrections, patterns and escalation. Attendance is a domain record, not inferred solely from a video join event.

6.12 Assessments and Item Bank

Outcome-aligned question bank, QTI import/export, metadata, difficulty, cognitive level, language, exposure history, variants, review status and item statistics. Assessment types include quizzes, exams, assignments, observations, portfolios, oral tasks, practicals, peer review and group work.

6.13 Gradebook

Weighted categories, points, percentages, competency/mastery, pass/fail, letter grades, dropped scores, exemptions, incomplete status, late penalties, resits, moderation, ratification and release. Calculations are deterministic, versioned and explainable to staff and learner.

6.14 Integrity and Examination Operations

Identity assurance, secure-browser integration, question randomisation, restricted navigation, proctor-provider integration, incident logs, candidate declarations and evidence review. Veza must not claim that webcam monitoring proves cheating; it records signals for human review and due process.

6.15 Progress and Mastery

Completion, time-on-task signals, outcome mastery, pace, attendance, assessment trajectory and engagement. Learners see transparent progress and recommended actions; staff see actionable exceptions. Time-on-page is treated as a weak signal, never learning proof.

6.16 Learner Support and Case Management

Advising, accommodations, tutoring, wellbeing referral, technical support and academic intervention. Cases have type, sensitivity, owner, SLA, tasks, communications, outcomes and field-level access rules.

6.17 Content and Media Library

Tenant-wide assets with ownership, licence, alt text, transcript, language, tags, retention, usage references, duplicates, virus scan and processing status. Replacing an asset creates a new version and warns where it is used.

6.18 Credentials and Records

Certificates, statements of results, transcripts, Open Badges 3.0 and CLR 2.0. Credentials are verifiable, revocable where policy allows, expiry-aware and tied to immutable achievement evidence.

6.19 Commerce

Public catalogue, products, bundles, coupons, tax, invoices, payment plans, sponsored seats, refunds and revenue reporting. Academic enrolment is created only after the relevant commercial and eligibility conditions pass.

6.20 Analytics and BI

Operational dashboards, cohort analysis, learning analytics, assessment quality, instructor workload, programme health, accreditation evidence and exports. Every metric has a definition, population, freshness, owner and drill-through path.

6.21 Integration Hub

SSO, SCIM, OneRoster, LTI Advantage, QTI, Common Cartridge, webhooks, REST/GraphQL APIs, SIS/HRIS/CRM connectors, video, storage, plagiarism, payment and messaging providers. Includes mapping, retries, dead-letter queues, replay and reconciliation.

6.22 AI Assist

Optional tenant-governed capabilities: lesson drafting, question suggestions, rubric drafting, summarisation, translation, feedback assistance, semantic search and learner tutor. AI output is labelled, source-grounded where relevant, reviewable, rate-limited and excluded from final high-stakes decisions without authorised human approval.

7. Assessment architecture

7.1 Assessment object model

Entity	Responsibility
Assessment blueprint	Purpose, outcomes, weight, cognitive distribution, item types, duration and security level.
Assessment definition	Instructions, settings, rubric, question selection rules, attempts, timing and accommodations.
Delivery instance	Concrete candidate attempt with resolved questions, timing, environment and integrity configuration.
Response	Immutable candidate response events and final response state.
Submission	Attempt closure, receipt, files, checksum and late/incident metadata.
Evaluation	Marker decisions, rubric scores, annotations, feedback and flags.
Moderation	Sample, reviewer decisions, variance and resolution.
Result	Calculated score, grade, outcome evidence, status and publication state.
Appeal	Grounds, evidence, reviewers, decision and any authorised result correction.

7.2 Grade calculation safeguards

- Calculation rules are stored as versioned expressions and evaluated by a deterministic service; spreadsheet-like flexibility is provided through constrained, testable functions rather than arbitrary code.

- Every gradebook displays the calculation explanation and identifies excluded, missing, excused, late and overridden items.
- Publishing creates a result snapshot. Subsequent recalculation cannot silently rewrite the published record.
- Rounding is configured once per grade scheme and applied only at defined boundaries.
- Bulk changes run in preview mode, show affected learners and require a reason before commit.

8. Learning analytics and data ethics

Veza should implement an event vocabulary compatible with Caliper and optionally xAPI while maintaining a canonical internal event envelope. Raw events feed an immutable stream; derived metrics are calculated separately. This prevents dashboard logic from becoming the system of record and enables recalculation when definitions improve.

Layer	Data	Retention/use
Operational state	Current enrolments, grades, attendance, submissions	Primary transactional record governed by institutional retention.
Event stream	Viewed, started, submitted, attended, messaged, graded events	Append-only, tenant-scoped; pseudonymised for product analytics where possible.
Metric store	Completion, pace, mastery, risk factors, item statistics	Versioned metric definitions with freshness and lineage.
Intervention record	Human review, action, follow-up and outcome	Sensitive institutional record with narrow access.
Product telemetry	Performance, errors, feature usage	No learner content; used for reliability and product improvement under contract.

A risk score must never be presented as an objective truth. The UI must show contributing signals, missing data, last update and recommended review. Institutions configure thresholds and intervention policy. Veza audits false-positive patterns across protected or sensitive cohorts where lawful and appropriate.

9. Information architecture and UX

9.1 Global shell

Region	Behaviour
Top bar	Institution switcher, global search/command palette, create button, notifications, help and account. It remains visually quiet.
Primary sidebar	Stable applications: Home, People, Learning, Studio, Assess, Calendar, Communicate, Insights and Admin. Labels adapt to role and tenant modules.
Context rail	Appears inside a course, programme or administrative object and contains that object's navigation. It is not another global sidebar.

Main canvas	Task workspace. Uses bento groupings only when multiple related signals/actions must be scanned together.
Inspector panel	Right-side contextual properties, comments, learner details or settings. It opens on demand and preserves the main context.
Command palette	Searches people, courses, actions, settings and help; keyboard-first for expert operators.

9.2 Role-specific navigation

Role	Default landing	Primary navigation emphasis
Learner	Continue learning	Today, My Learning, Calendar, Messages, Progress, Help
Instructor	Teaching command centre	Today, Classes, Studio, Assess, Calendar, Learners, Messages
Registrar	Operations overview	People, Enrolment, Academic Structure, Timetable, Records, Reports
Executive	Institution pulse	Outcomes, enrolment, engagement, risks, compliance, finance
Platform operator	Service operations	Tenants, provisioning, plans, integrations, incidents, usage

9.3 Bento principles

- A card exists only when it encapsulates a distinct decision, metric or workflow. Long record lists remain tables or timelines, not dozens of cards.
- Card size reflects information importance and interaction complexity, not visual variety.
- Bento layouts are responsive: desktop grid, tablet two-column and mobile ordered stack. Priority order is explicitly defined, not left to CSS wrapping.
- Whitespace separates conceptual groups; borders are low-contrast; elevation is reserved for transient layers and draggable objects.
- Empty states explain the operational meaning, offer one primary action and link to relevant setup—not generic illustrations.

9.4 Visual identity and colour system

Recommended brand: Veza Learning Cloud. “Veza” means reveal/show in isiZulu, which aligns naturally with making learning, progress and institutional operations visible. “Learning Cloud” is broad enough for the product suite, while Veza Studio, Veza Assess, Veza Live and Veza Insights form clear module names.

Token	Value	Use
Ink 950	#0B1220	Primary text, navigation and high-emphasis controls
Cloud 0	#FFFFFF	Primary surfaces

Cloud 50	#F6F8FB	Application background and grouped work areas
Cloud 200	#DCE3EC	Dividers and input borders
Veza Blue 600	#2563EB	Primary actions, focus, active navigation and links
Veza Teal 600	#0D9488	Learning/progress accent, secondary data and success-adjacent brand moments
Amber 500	#F59E0B	Attention and pending states, never general decoration
Red 600	#DC2626	Destructive actions and critical risk
Green 600	#16A34A	Confirmed success and completion

Colour is semantic and restrained. Institution branding may set logo, type accent and limited theme tokens, but Veza retains accessible system colours for statuses, focus and validation. Tenant colours are automatically checked for contrast and are never allowed to redefine red/green status meaning.

9.5 Typography and density

- Use a highly legible variable sans such as Inter or Geist for the product UI; use one family across tenant brands unless an enterprise brand font passes performance and accessibility checks.
- Default body 14-16 px depending on surface; operational tables can use 13-14 px with generous row height and user-selectable comfortable/compact density.
- Headings communicate hierarchy, not decoration. Avoid oversized marketing typography inside the authenticated product.
- Numerical dashboards use tabular figures and align units; never rely on colour alone to communicate change.

10. Technical architecture

10.1 Recommended stack

Layer	Recommendation	Why it fits this platform
Web applications	Next.js 16+, React, TypeScript, Tailwind, Radix primitives, TanStack Query/Table, React Hook Form, Zod	Supports dense admin surfaces, SSR/edge delivery for public catalogues, strong component ecosystem and shared type-safe UI.
Mobile	React Native/Expo after web core; offline learner cache and push notifications	Shares TypeScript/domain packages while enabling downloads, camera evidence, push and better constrained-network behaviour.
Backend	NestJS on Node.js with modular bounded contexts; REST/OpenAPI externally, internal events; GraphQL only for selected aggregate reads	NestJS provides explicit modules, guards, validation and testability. REST aligns well with integrations; selective GraphQL avoids universal complexity.
Primary data	PostgreSQL 17+ via AWS RDS/Aurora;	Education data is relational and

	tenant_id on pooled records; row-level policies as defence in depth	transactional. PostgreSQL supports complex reporting, constraints, JSON where appropriate and mature tooling.
Cache/coordination	Redis/ElastiCache	Sessions, rate limits, distributed locks, queues and short-lived calculated views.
Object/media	S3 with tenant prefixes and KMS, CloudFront, MediaConvert/Elemental or provider abstraction	Durable media, signed access, lifecycle tiers, transcoding and global delivery.
Search	OpenSearch initially for global search; vector capability isolated behind search service	Cross-domain search, filtering and audit-friendly indexing. Vector search can evolve without contaminating source records.
Events/jobs	EventBridge + SQS; outbox pattern from PostgreSQL; Step Functions for long workflows	Reliable integration, retries, scheduled work and auditable multi-step processes.
Identity	Amazon Cognito or Auth0/WorkOS abstraction; OIDC/SAML, SCIM, MFA, passkeys; institution federation	Institution customers require federation and lifecycle automation. Identity remains a replaceable platform capability.
Observability	OpenTelemetry, CloudWatch, managed tracing/log pipeline, product analytics separated from audit logs	Tenant-aware diagnostics and SLOs without mixing operational telemetry with academic records.
Infrastructure	AWS CDK or Terraform, GitHub Actions, container workloads on ECS Fargate initially	Repeatable environments and less operational overhead than Kubernetes before scale requires it.

10.2 Bounded contexts

Context	Owns	Must not own
Tenant & Entitlements	Its entities, commands, invariants and emitted domain events	Another context's tables or business rules
Identity & Access	Its entities, commands, invariants and emitted domain events	Another context's tables or business rules
Institution Structure	Its entities, commands, invariants and emitted domain events	Another context's tables or business rules
People & Relationships	Its entities, commands, invariants and emitted domain events	Another context's tables or business rules
Admissions	Its entities, commands, invariants and emitted domain events	Another context's tables or business rules
Academic Catalogue	Its entities, commands, invariants and emitted domain events	Another context's tables or business rules

Curriculum	Its entities, commands, invariants and emitted domain events	Another context's tables or business rules
Course Delivery	Its entities, commands, invariants and emitted domain events	Another context's tables or business rules
Content & Media	Its entities, commands, invariants and emitted domain events	Another context's tables or business rules
Scheduling	Its entities, commands, invariants and emitted domain events	Another context's tables or business rules
Communication	Its entities, commands, invariants and emitted domain events	Another context's tables or business rules
Attendance	Its entities, commands, invariants and emitted domain events	Another context's tables or business rules
Assessment	Its entities, commands, invariants and emitted domain events	Another context's tables or business rules
Gradebook & Results	Its entities, commands, invariants and emitted domain events	Another context's tables or business rules
Learner Support	Its entities, commands, invariants and emitted domain events	Another context's tables or business rules
Credentials	Its entities, commands, invariants and emitted domain events	Another context's tables or business rules
Commerce	Its entities, commands, invariants and emitted domain events	Another context's tables or business rules
Analytics	Its entities, commands, invariants and emitted domain events	Another context's tables or business rules
Integrations	Its entities, commands, invariants and emitted domain events	Another context's tables or business rules
Audit & Compliance	Its entities, commands, invariants and emitted domain events	Another context's tables or business rules

Modules communicate through application services and domain events, not direct table writes. A modular monolith can keep these modules in one deployment initially, but database schemas, ownership rules and contracts are enforced from the start.

10.3 Request and event flow

Flow	Sequence
Authenticated command	Web/mobile -> edge/WAF -> API -> session validation -> tenant context -> authorisation policy -> domain service -> transaction + outbox -> response -> asynchronous consumers.
File upload	Client requests upload intent -> policy/size/type validation ->

	signed multipart upload -> quarantine bucket -> malware scan -> metadata extraction -> approved bucket -> media processing -> asset-ready event.
Analytics event	Domain/UI event -> event gateway -> schema validation -> stream/archive -> transformation -> metric store -> dashboards. Critical academic state is never reconstructed solely from analytics events.
Integration sync	Scheduled/webhook trigger -> connector -> mapping/validation -> idempotency check -> domain commands -> result ledger -> retries/dead-letter -> reconciliation dashboard.

10.4 Multi-tenant data controls

- Tenant context is embedded in the authenticated token/session and propagated through request context, database session, cache keys, object paths, events, logs and metrics.
- Every pooled table includes tenant_id in primary/unique indexes where relevant. Database row-level security provides defence in depth, but application authorisation remains mandatory.
- Background jobs carry signed tenant context and are rejected if the target record does not match it.
- Cross-tenant operator queries use a separate control-plane path and privileged identity, never ordinary tenant APIs with a missing filter.
- Automated isolation tests create two tenants and continuously attempt horizontal access across every resource API.

10.5 Core data entities

Entity	Key responsibility
Tenant	plan, residency, entitlements, branding, identity config
Institution	legal/academic profile and hierarchy
User	global identity
Membership	user-to-tenant relationship and status
RoleAssignment	role, scope, conditions, validity
PersonProfile	tenant-controlled person data
Relationship	guardian, sponsor, manager, advisor
AcademicPeriod	nested time structure
Programme	requirements and outcomes
CourseOffering	versioned curriculum definition
CourseRun	delivery instance
Section	class/cohort grouping
Enrolment	person participation and status

ContentItem	versioned learning object
Activity	learner action requirement
Session	scheduled teaching event
AttendanceRecord	official presence status/evidence
Assessment	definition and blueprint
Attempt	candidate delivery instance
Submission	immutable submitted evidence
Evaluation	marker decision
Result	published academic result
Competency	outcome framework node
EvidenceLink	result/activity-to-competency evidence
SupportCase	intervention/advising workflow
Credential	issued achievement record
AuditEvent	immutable privileged/action record

11. Interoperability strategy

Standard	Veza use
LTI Advantage / LTI 1.3	Secure launch of external tools, deep linking, names/roles, assignment and grade services. Veza acts as platform first and can later expose Veza modules as tools.
OneRoster 1.2	Roster, class, course and grade exchange with school information systems using CSV and REST profiles.
QTI 3	Portable assessment items/tests and results; preserve unsupported extensions rather than silently flattening them.
Common Cartridge	Import/export course packages and resources with a transparent compatibility report.
Caliper Analytics	Common vocabulary for learning activity and tool usage; map canonical Veza events to Caliper profiles.
xAPI	Optional learning record interoperability for experiences outside the LMS and enterprise training ecosystems.

Open Badges 3.0 / CLR 2.0	Verifiable, portable achievements and longitudinal learner records.
OIDC/SAML 2.0	Federated authentication for institutions.
SCIM 2.0	Automated user and group provisioning/deprovisioning.
iCalendar	Calendar subscription and event exchange.

12. Security, privacy and compliance

12.1 Security baseline

- Adopt OWASP ASVS 5.0 as the web application verification baseline and MASVS for native mobile applications.
- Use secure-by-default tenant templates, MFA for privileged roles, passkey support, risk-based step-up, session/device management and rapid deprovisioning.
- Encrypt in transit and at rest; use tenant-aware KMS context and separate secrets from application configuration.
- Centralise immutable audit events for authentication, role changes, exports, grade actions, record access, policy changes and support elevation.
- Perform dependency scanning, SAST, secret scanning, IaC scanning, container scanning, DAST and regular penetration tests. High-risk changes require threat modelling.
- Backups are encrypted, point-in-time capable and restore-tested. Disaster recovery objectives are explicit per tier.

12.2 POPIA operating model

For most institution deployments, the institution determines the purpose and means of processing learner information and is the responsible party; Veza acts primarily as operator under a written agreement. Veza may be a responsible party for its own billing, account, security and product-operation data. The contract and privacy notices must separate these roles clearly.

Control	Implementation
Purpose and minimisation	Data field catalogue records purpose, lawful basis/authority, sensitivity, owner and retention. Optional fields are disabled unless needed.
Operator agreement	Written processing terms, confidentiality, security safeguards, subprocessors, assistance, deletion/return and breach notification.
Children and guardians	Age/authority rules, guardian relationships, consent where required, restricted communications and age-appropriate notices.
Data subject requests	Search, review, export, correction, restriction and deletion workflow with legal/academic record exceptions.
Security compromise	Tenant-aware incident workflow, evidence preservation, responsible-party notification and support for regulator/data-subject reporting.
Cross-border processing	Per-tenant residency and subprocessor register; contractual and legal transfer controls before enabling a region/provider.

Retention	Policy engine by record class, lifecycle holds, archive and verified destruction.
-----------	---

12.3 Accessibility

Target WCAG 2.2 Level AA for the product and for content generated in Veza Studio. Accessibility is a product invariant, not a final audit. Components, keyboard operation, focus, contrast, reflow, captions, transcripts, form errors, timing controls and assessment accommodations are tested continuously.

- Authoring checker prevents or warns about missing alternative text, skipped headings, inaccessible link text, colour-only meaning, low contrast, uncaptioned media and inaccessible attachments.
- Assessment delivery supports extended time, pauses where policy allows, alternative presentation, screen readers, keyboard navigation, zoom, language alternatives and per-candidate settings.
- Accessibility metadata travels with content during import/export where standards support it.
- No drag-only interaction; every canvas action has keyboard and form alternatives.

13. Reliability and operations

Area	Target design
Availability	99.9% core tier initially; higher contractual tier through dedicated architecture. Status page and transparent incident communication.
Performance	p95 interactive API under 500 ms for normal operations; large reports are asynchronous. Learner critical path optimised for low-bandwidth networks.
RPO/RTO	Shared tier target RPO <= 15 minutes, RTO <= 4 hours; protected/sovereign options improve by contract.
Noisy neighbour	Per-tenant rate, concurrency, queue and storage quotas; expensive jobs isolated; fairness scheduling.
Release safety	Feature flags, tenant rings, backwards-compatible migrations, canaries, rollback and contract tests.
Support	In-product diagnostics bundle, tenant health view, runbooks, case linking and privacy-preserving logs.
Archival	Course closure package includes content version, roster, grade schema, results, moderation, key communications and audit references.

14. Quality assurance and flow validation

14.1 Required test layers

Layer	Coverage
Domain unit tests	Progression, enrolment, date windows, grade calculations,

	permissions, completion and retention.
Property-based tests	Grade formula invariants, timetable constraints, randomised assessment assembly and tenant isolation.
Contract tests	All bounded-context APIs, integration adapters and event schemas.
Workflow tests	Complete admission, course approval, teaching, submission, marking, moderation, release, appeal and credential flows.
Permission matrix tests	Every permission x role x scope x condition, including explicit denial and expired assignment.
Accessibility tests	Automated component checks plus keyboard, screen-reader and assessment usability testing.
Visual regression	Design-system components and critical pages at supported breakpoints and density modes.
Load tests	Course launch spikes, exam start/submit, mass enrolment, grade release, video access and notification storms.
Chaos/recovery	Queue delay, provider outage, partial media processing, database failover and restore rehearsal.
Migration reconciliation	Counts, checksums, sampled records, unresolved mappings and signed acceptance report.

14.2 Gap tests for critical workflows

Scenario	Required handling
Instructor leaves mid-course	Reassign ownership, preserve authored content, transfer pending marking and notify affected staff.
Learner changes class	Effective-dated enrolment; attendance/grades remain attributed; future timetable and groups update.
Course blueprint is updated after runs exist	Existing runs remain on pinned version; manager receives upgrade diff and chooses what to merge.
Video provider is unavailable	Session status, fallback link/provider, learner notice and attendance contingency are available without losing the event.
Exam connection drops	Local/periodic autosave, resumable session under policy, incident log and no duplicate attempt.
Grade formula changes after marks exist	Preview impact, approval, versioned calculation, no silent rewrite of published results.
Duplicate learner identities	Merge workflow with evidence, conflict handling, immutable source IDs and rollback plan.

Tenant contract ends	Export package, retention/legal holds, access shutdown, deletion schedule and certificate of destruction.
Integration sends stale update	Source version/timestamp and idempotency rules reject or queue conflict; reconciliation UI resolves.
AI generates unsafe or incorrect content	Labelled draft, source display, policy filters, human approval and incident feedback; never auto-publish high-stakes material.

15. Delivery roadmap

Phase	Outcome	Scope
0. Foundations	A trustworthy SaaS control plane and design system	Tenancy, identity, RBAC/policy, audit, institution setup, feature entitlements, design system, CI/CD, observability and security baseline.
1. Core Learning	Institution can create, enrol and teach	People, academic structure, course blueprint/run, Studio basics, learner course room, calendar, announcements, assignments, basic gradebook and progress.
2. Institutional Operations	Institution can run repeatable academic operations	Admissions, bulk roster, attendance, timetable, advanced roles, guardian/sponsor, support cases, reports and SSO/SCIM.
3. Assessment Rigour	Institution can conduct credible assessment	Item bank, QTI, exams, rubrics, moderation, grade ratification, appeals and accommodations.
4. Ecosystem and Credentials	Platform is portable and extensible	LTI, OneRoster, Common Cartridge, integration hub, Open Badges, CLR, APIs and webhooks.
5. Intelligence and Scale	Actionable analytics and advanced commercial model	Caliper/xAPI mapping, intervention analytics, commerce, marketplace, AI Assist, native app and dedicated tiers.

16. MVP boundary and non-negotiables

The first commercial release should not attempt every module. It must, however, establish the architecture that prevents later rework. The MVP is sellable to a training provider or small institution and includes tenant provisioning, branding, identity, scoped RBAC, people/imports, academic periods, course templates and runs, Studio block authoring, learner delivery, calendar, announcements, assignments, rubrics, gradebook, progress, certificates, audit, exports and core analytics.

The following are non-negotiable even in MVP: tenant isolation tests, accessible component library, immutable submissions, versioned course content, explainable grade calculations, data export, audit logs, backup restore tests, role scope, consent-aware guardian relationships and integration-ready identifiers.

17. Product principles that prevent “AI slop”

- Every feature must identify the institutional decision or job it supports, its owner, its record of truth and its failure states.
- No dashboard metric ships without definition, denominator, freshness, owner and drill-through.
- No workflow state is represented only by a coloured badge; state transitions, permissions and audit consequences are explicit.
- No “smart” automation silently changes academic records. Automation proposes, validates or routes; authorised humans approve consequential decisions.
- No decorative bento proliferation. Dense work uses tables, timelines, canvases and inspectors where those are the correct interaction models.
- No institution-specific custom fork. Variance is implemented through configuration, policy, terminology, workflow and extension points.
- No integration is considered complete without retries, idempotency, reconciliation, mapping visibility and support diagnostics.

18. Standards and validation references

Reference	URL	Use in this architecture
1EdTech Interoperability Standards	https://www.1edtech.org/specifications	Source catalogue for LTI, OneRoster, QTI, Caliper and related standards.
Learning Tools Interoperability (LTI)	https://www.1edtech.org/standards/lti	LTI 1.3/LTI Advantage security and service model.
OneRoster	https://www.1edtech.org/standards/oneroster	Roster, course and grade exchange; current listed version 1.2.
Question & Test Interoperability	https://www.1edtech.org/standards/qti	Portable assessment items, tests and results.
Common Cartridge	https://www.1edtech.org/standards/cc	Portable learning-resource and course packages.
Caliper Analytics	https://www.1edtech.org/standards/caliper	Common learning activity and usage analytics vocabulary.
Open Badges	https://www.1edtech.org/standards/open-badges	Portable verifiable achievements; Open Badges 3.0.
Comprehensive Learner Record	https://www.1edtech.org/standards/clr	Learner-controlled longitudinal achievement records; CLR 2.0.
WCAG 2.2	https://www.w3.org/TR/WCAG22/	W3C accessibility recommendation and Level AA target.
AWS SaaS Lens - Tenant Isolation	https://docs.aws.amazon.com/wellarchitected/latest/saas-lens/tenant-isolation.html	Tenant isolation and SaaS identity principles.
OWASP ASVS	https://owasp.org/www-project-application-security-verification-standard/	Application security requirements and verification baseline; ASVS 5.0 released in 2025.

Information Regulator - POPIA	https://info regulator.org.za/popia/	South African privacy regulator guidance and breach responsibilities.
-------------------------------	---	---

19. Final recommendation

Build and market the product as Veza Learning Cloud. Use Veza Studio as the flagship creation environment and name the remaining modules by their job: Veza Admin, Veza Teach, Veza Assess, Veza Live, Veza Insights and Veza Credentials. The platform should feel coherent because every module shares the institutional graph, permissions, events, search, audit and design system—not because every screen uses the same cards.

The decisive architectural choice is to treat institution operations, learning design, delivery and evidence as one connected lifecycle while preserving standards-based portability. That gives Veza a defensible position between conventional LMS products, fragmented education administration suites and generic content-authoring tools.